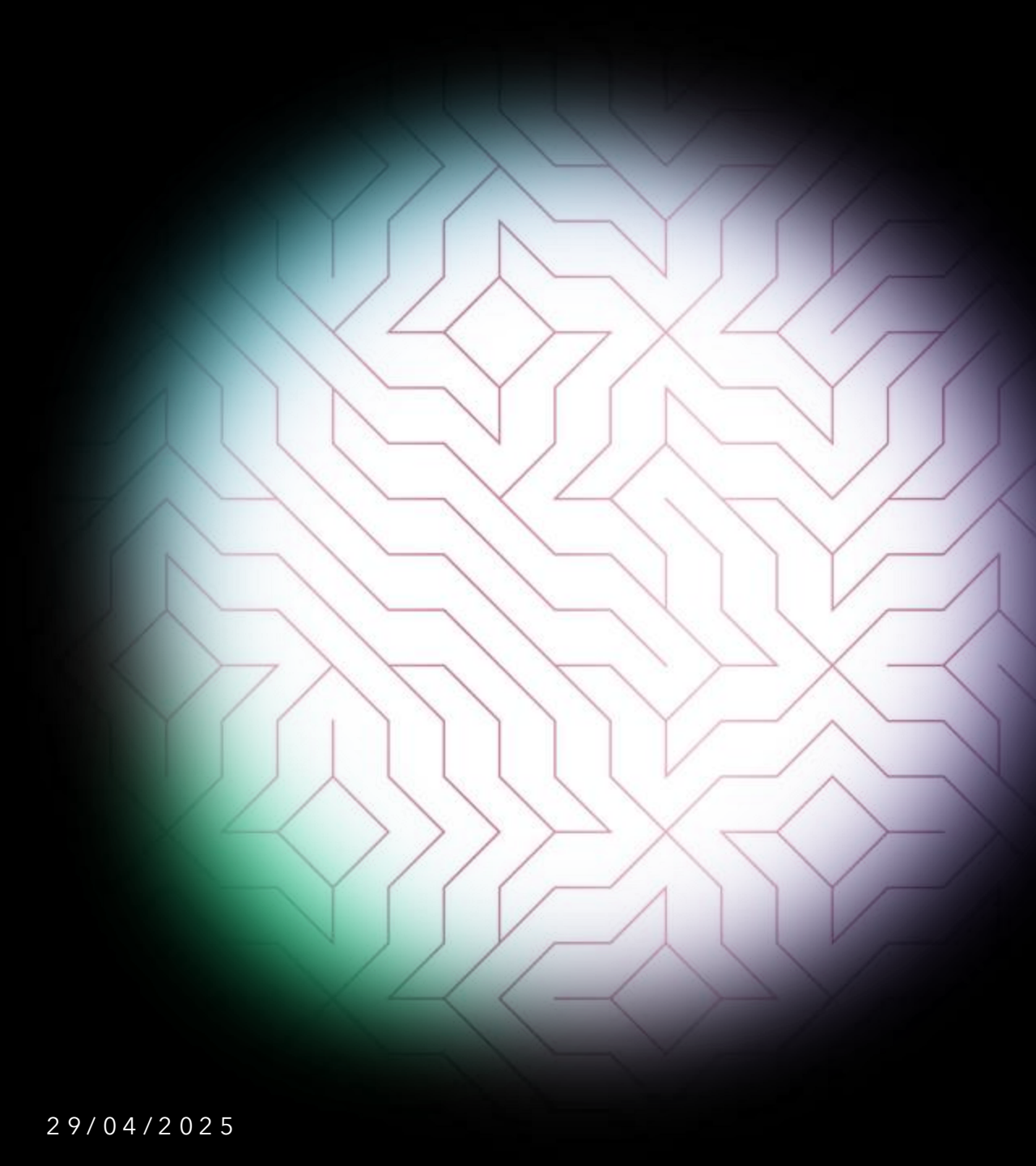




Cambridge MIND Hackathon

BLUE TEAM

Marcella Montagnese, Clara Pecci,
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Issues



Pre-PhD student

Challenge of basic understanding –
how to extract basic information about
the data without the use of specific
software and coding knowledge



Issues



Being able to quickly access, understand
and extract key information from open
access datasets

PhD & Postdoc

Pre-PhD student

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PI/Supervisor

Understanding the high-level overview of data before pre-processing across different modalities in a more standardised manner



Issues



Being able to quickly access, understand and extract key information from open access datasets

Ease of use

PhD & Postdoc

High-level overview

Pre-PhD student

*Efficiency,
reproducibility*

PI/Supervisor

Challenge of basic understanding – how to extract basic information about the data without the use of specific software and coding knowledge

Understanding the high-level overview of data before pre-processing across different modalities in a more standardised manner

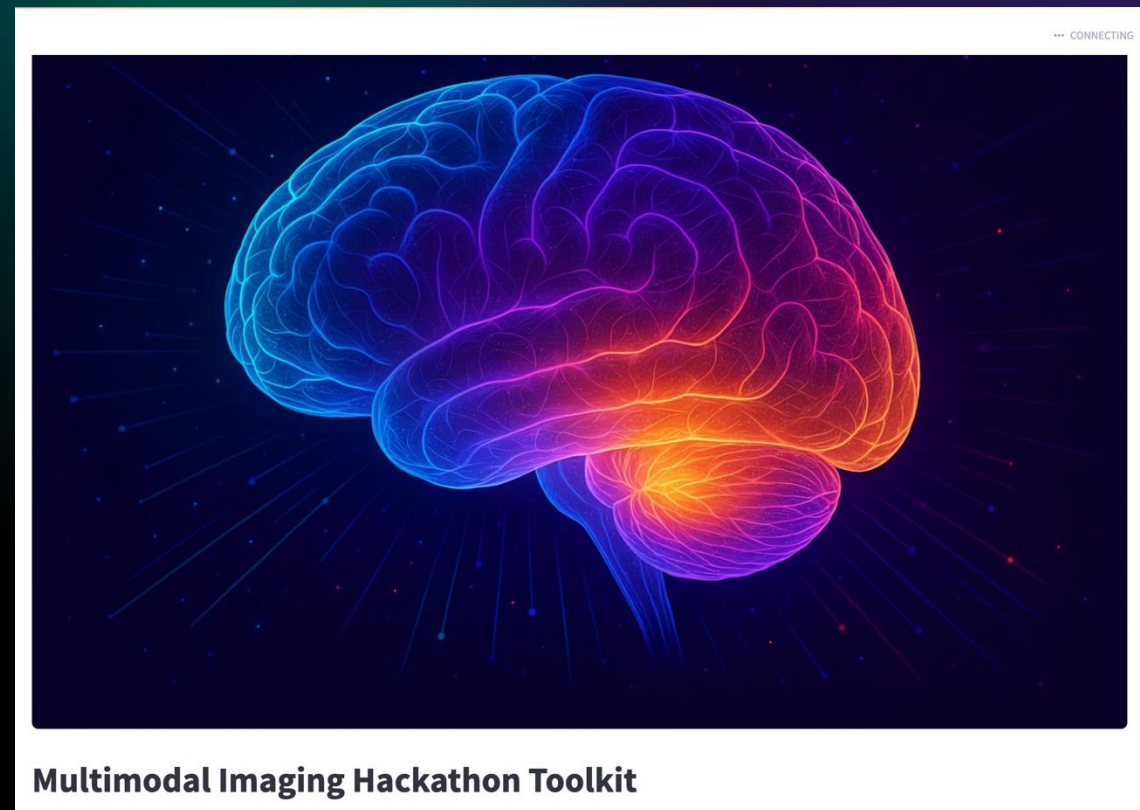


Solution

Multimodal Imaging Hackathon Toolkit

Easy-to-use toolkit for first look at neuroimaging data across modalities

- Load and inspect neuroimaging data
- Interact with AI to interpret
- All within an easy web interface – no coding needed



Workflow



1. Upload file or folder with data



2. Extracts relevant information



3. Outputs PDF file & plots



4. Chat assistant for interpretation

Workflow



1. Upload file or folder with data

Drag and drop to Git

Supported files = (".edf",
".json", ".tsv", ".nii", ".nii.gz")

Download from Google Drive

Enter Google Drive file URL or ID

Download and Extract

Select Imaging Modality and Files

Select Modality

fNIRS (.snirf / .nirs) ▼

Select fNIRS .snirf/.nirs file

No fNIRS files found ▼

Select accompanying .json file

./data/func/sub-15_task-genMu... ▼

Workflow



2. Extracts relevant information

- Modality (EEG, fMRI)
- Information on imaging process
- Pre-processing guidance (pipelines, software, toolbox suggestion)
- Quality control metrics (signal-to-noise ratio & interpretation suggestion)
- Relevant warnings & recommendations (potential pre-processing issues)
- Visualisation plots & guidance on interpretation (e.g., EEG signal overview and its meaning, fMRI slice)

Workflow



Multimodal Imaging Hackathon Toolkit

✓ Selected both .nii.gz and .json files from repo

BOLD NIfTI Information

Shape: (64, 64, 37, 277)

Affine:

```
[[-2.99999976e+00  3.82781029e-04  1.65611506e-03  9.59579391e+01]
 [ 2.49385834e-04  2.98517942e+00 -3.72288108e-01 -5.40791092e+01]
 [ 1.35636330e-03  2.97830403e-01  3.73147416e+00 -8.11782990e+01]
 [ 0.00000000e+00  0.00000000e+00  0.00000000e+00  1.00000000e+00]]
```

Metadata from JSON

```
{
  "RepetitionTime" : 2
  "EchoTime" : 0.03
  "FlipAngle" : 90
  "TaskName" : "genMusic01"
}
```

Workflow



3. Outputs PDF file & plots

fMRI

This is a fMRI image. To preprocess these images we recommend installing [nibabel](#) library.

Quality Control

Additionally, to automate quality control (QC), you may consider [nilearn](#) and [fMRIPrep](#) packages.

Warnings/Recommendations

The signal-to-noise ratio (SNR) is 40. Acceptable but more susceptible to noise.

Visualization

Plot option 1: Mean BOLD signal per voxel; Plot option 2: Voxel time series representing the timeseries of a voxel at a given (x,y,z) coordinates

Workflow



3. Outputs PDF file & plots

EEG

This is a EEG image. To preprocess these images we recommend installing [MNE](#) library.

Quality Control

Additionally, to automate quality control (QC), you may consider [PyEEG](#) package.

Warnings/Recommendations

The signal-to-noise ratio (SNR) is 40. Acceptable but more susceptible to noise.

Visualization

Plot option 1: Plot time series of EEG channels

Workflow

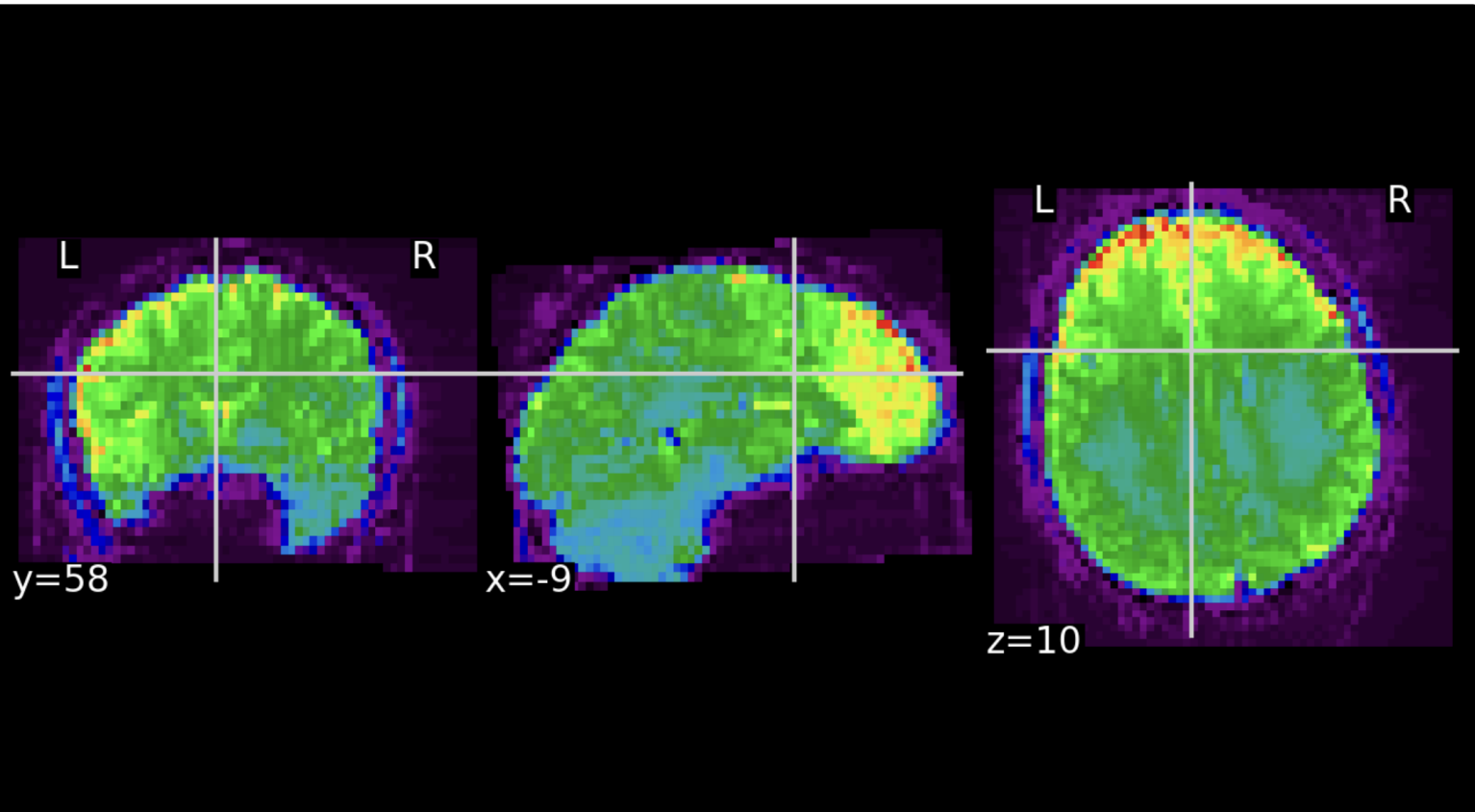


Choose an Action on the Imaging Data

Select an analysis to perform:

Plot Mean BOLD Signal

Run Selected Analysis



Workflow



4. Chat assistant for interpretation

Ask something about your neuroimaging data:

what type of signal is my data capturing and what are the stimulus timings?

Send

Assistant Response:

Based on the provided metadata and the shape of the NIfTI data, it seems like you are working with functional MRI (fMRI) data. The shape of (64, 64, 37, 277) indicates that the data consists of 277 volumes, each with a 64x64 voxel grid across 37 slices.

The metadata also reveals important details about the fMRI acquisition:

- Repetition Time (TR): 2.0 seconds
- Echo Time (TE): 0.03 seconds
- Flip Angle: 90 degrees
- Task Name: "genMusic01"

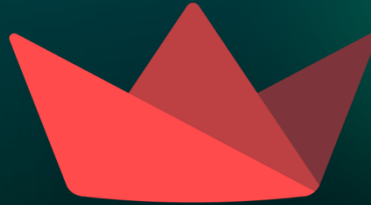


Run through...

Methods



Github
Codespaces



Streamlit

App created based on
tutorial/workshop on
LLM by Accelerated
Science Team



OpenAI



Multimodal integration in
one user-friendly platform



Benefits



Multimodal integration in
one user-friendly platform



- **Improves efficiency** (no need to build ad-hoc pipelines to integrate and understand cross modality data)
- **Simplifies initial process** (useful for students new to these methods, proficient users looking for efficiency and PIs looking for more high-level insight into data)

Benefits

Novelty

Multimodal integration in one user-friendly platform

Impact

- **Improves efficiency** (no need to build ad-hoc pipelines to integrate and understand cross modality data)
- **Simplifies initial process** (useful for students new to these methods, proficient users looking for efficiency and PIs looking for more high-level insight into data)

Benefits

Refinement

- **Plug & play setup** (works with any data loaded onto Git),
- **Works standalone** (Github Codespaces can use any web browser)
- **Future proof** (BIDS format and use of Git format for open-source collaboration)

Novelty

Multimodal integration in one user-friendly platform

Impact

- **Improves efficiency** (no need to build ad-hoc pipelines to integrate and understand cross modality data)
- **Simplifies initial process** (useful for students new to these methods, proficient users looking for efficiency and PIs looking for more high-level insight into data)

Benefits

Rigour

Promotes open science principles including:

- **Accessibility** (free, open access, easy-to-use, does not require software)
- **Transparency & standardisation** (each project produces a standardised README file explaining basic information about the data)

Refinement

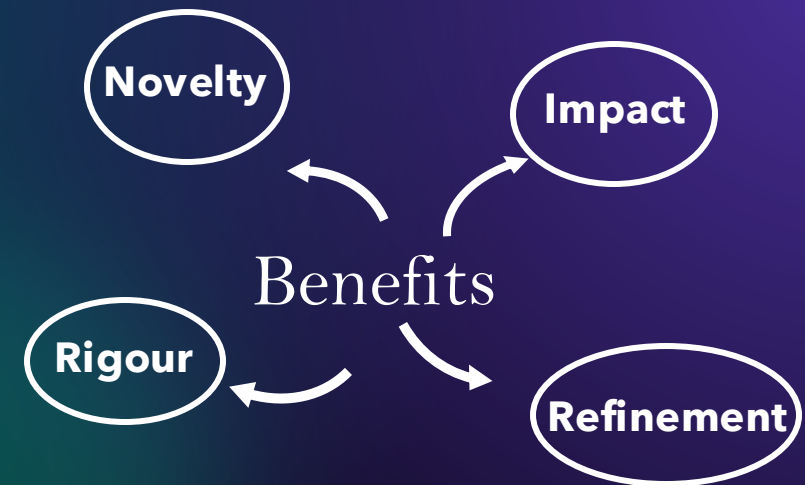
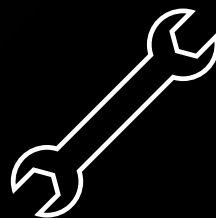
- **Plug & play setup** (works with any data loaded onto Git),
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Thank you!

Multimodal Imaging Hackathon Toolkit

Easy-to-use toolkit for first look at neuroimaging data across modalities –

- Useful from beginner to advanced
- Useful for everyday data exploration to your next Hackathon!



Marcella Montagnese, Clara Pecci, Natalia Garcia-San-Martin, Sara De Felice, Kirsten Thomas